L03: Experimental Design, Blocking, Balancing

S&DS 6165: Topics in Causal Inference

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Lecture brief: View the source `brief.md` · How these notes are made

Lecture outline

- Use a grouped-assignment example to show why marginal treatment probabilities do not determine precision.
- Develop blocking and matched pairs as design-stage uses of pretreatment covariates.
- Contrast design-stage balance with regression and AIPW corrections made during analysis.
- Treat rerandomization as a complete design whose balance restrictions must also govern inference.

1 Ten coin flips disguised as one thousand

An experimental design is a probability distribution over complete assignment vectors, not merely a list of unit-level treatment probabilities. The distinction matters whenever assignments are dependent. As an extreme example, suppose $N = 1000$ units each have marginal treatment probability $1/2$. Partition them into ten groups of one hundred and flip one fair coin per group. Every unit in a group receives the same assignment.

The marginal probabilities match i.i.d. Bernoulli assignment, but the design contains only ten independent random decisions. Let $g = 1, \ldots, 10$ index the groups, let $\mathcal{J}_g$ be the set of units in group g, and let $A_g \in \{0, 1\}$ be the result of its coin flip. Thus $Z_i = A_g$ whenever $i \in \mathcal{J}_g$. With treatment probability $1/2$, the observable HT estimator (Horvitz and Thompson 1952) can be written first using observed outcomes and then, by consistency, using the corresponding potential outcomes:

$$\hat{\tau}_{\mathrm{HT}} = \frac{1}{N}\sum_{i=1}^{N}\left\{2Z_iY_i^{\mathrm{obs}} - 2(1-Z_i)Y_i^{\mathrm{obs}}\right\} = \frac{1}{N}\sum_{i=1}^{N}\left\{2Z_iY_i(1) - 2(1-Z_i)Y_i(0)\right\}.$$

The second expression may look as though it uses both potential outcomes, but its assignment indicators select only the observed one: the $Y_i(0)$ term vanishes for a treated unit and the $Y_i(1)$ term vanishes for a control unit. Subtracting the fixed estimand and collecting units that share a coin gives the exact error representation

$$\hat{\tau}_{\mathrm{HT}} - \tau = \frac{1}{N}\sum_{g=1}^{10}(2A_g - 1)\sum_{i\in\mathcal{J}_g}\{Y_i(1) + Y_i(0)\}.$$

The inner sum is a fixed group total; all assignment randomness in that group is the single sign $2A_g - 1$. Independence of the ten signs therefore gives

$$\mathrm{Var}_Z(\hat{\tau}_{\mathrm{HT}}) = \frac{1}{N^2}\sum_{g=1}^{10}\left[\sum_{i\in\mathcal{J}_g}\{Y_i(1) + Y_i(0)\}\right]^2. \tag{1}$$

Under i.i.d. unit-level Bernoulli assignment, the corresponding variance is

$$\mathrm{Var}_Z(\hat{\tau}_{\mathrm{HT}}) = \frac{1}{N^2}\sum_{i=1}^{N}\{Y_i(1)+Y_i(0)\}^2. \tag{2}$$

The contrast between Equation 1 and Equation 2 is the location of the square. In the grouped design, outcomes are summed within a group and then squared, so the one hundred units sharing a coin move together. Under unit-level assignment, each unit contributes its own squared term.

Take the stark special case $Y_i(1) = a$ and $Y_i(0) = b$ for every unit. The grouped design has variance $(a+b)^2/10$. That is exactly the i.i.d. Bernoulli variance for **ten** units, not one thousand. I.i.d. Bernoulli assignment of all one thousand units has variance $(a+b)^2/1000$.

This exact equivalence is specific to HT in the example with constant potential outcomes. It partly reflects HT's random total weight: conditional on at least one treated and one control group, difference in means is exactly $a-b$ and has zero variance. With heterogeneous group totals, however, every estimator's assignment variation still originates in only ten coin flips. The example therefore combines two lessons from L02 and L03: normalization matters, and dependence determines the effective source of randomization.

The point is not that dependence is always harmful. The positive dependence in this example makes many units move together. Negative dependence, by contrast, can force treatment and control counts or covariate totals to balance and thereby improve precision. What determines these properties is the **joint distribution of the whole assignment vector Z**, not the marginal probabilities considered one unit at a time.

2 Blocking: randomize among comparable units

The grouped example used dependence badly: it made every unit in a group receive the same treatment. Blocking uses dependence for the opposite purpose. Imagine a student experiment in which prior achievement is strongly predictive of the outcome. A simple complete randomization can, by chance, place more high-achieving students in one arm. Before assigning treatment, we can divide students into groups with similar prior achievement and require both treatment arms to appear within every group. The comparison is then made among students who were similar before treatment.

Formally, suppose pretreatment covariates divide the finite population into H nonoverlapping **blocks**, also called strata. Let $\mathcal{I}_h$ denote the units in block h and $N_h = |\mathcal{I}_h|$ its size. We choose in advance to treat exactly N_{1h} units and control $N_{0h} = N_h - N_{1h}$ units in that block. **Stratified complete randomization**, or blocking, performs complete randomization independently within each $\mathcal{I}_h$. Thus units in different blocks are randomized separately, while assignments inside a block are negatively dependent because their treated count is fixed.

For arm $z \in \{0,1\}$, define the observed mean within block h by

$$\overline{Y}_{zh}^{\mathrm{obs}} = \frac{1}{N_{zh}}\sum_{i\in\mathcal{I}_h}\mathbf{1}\{Z_i = z\}Y_i^{\mathrm{obs}},$$

where $\mathbf{1}\{Z_i = z\}$ is the indicator that unit i was assigned to arm z, and N_{zh} is the fixed number assigned to that arm in block h. The within-block treatment-control contrast is

$$\hat{\tau}_h = \overline{Y}_{1h}^{\text{obs}} - \overline{Y}_{0h}^{\text{obs}}$$

Define the block-specific finite-population effect

$$\tau_h = \frac{1}{N_h} \sum_{i \in \mathcal{J}_h} \{Y_i(1) - Y_i(0)\}.$$

Complete randomization within the block makes $\hat{\tau}_h$ design-unbiased for τ_h. The population ATE is not an equally weighted average of the block effects unless all blocks have the same size: every unit, rather than every block, must receive equal weight. This yields

$$\hat{\tau}_{\text{block}} = \sum_{h=1}^{H} \frac{N_h}{N} \hat{\tau}_h. \tag{3}$$

The weights N_h/N are the fractions of experimental units in the blocks, so $\tau = \sum_h (N_h/N)\tau_h$ and Equation 3 is design-unbiased for τ. To describe its precision, let S_{zh}^2 be the finite-population variance of $Y_i(z)$ among units in block h, and let $S_{\tau h}^2$ be the within-block variance of the individual effects τ_i. Applying Neyman's finite-population variance calculation inside each block and then adding across the independently randomized blocks gives (Splawa-Neyman et al. 1990)

$$\text{Var}_Z(\hat{\tau}_{\text{block}}) = \sum_{h=1}^{H} \left(\frac{N_h}{N}\right)^2 \left(\frac{S_{1h}^2}{N_{1h}} + \frac{S_{0h}^2}{N_{0h}} - \frac{S_{\tau h}^2}{N_h}\right). \tag{4}$$

The factor $(N_h/N)^2$ appears because each random within-block contrast enters the overall estimator multiplied by N_h/N. As in L02, $S_{\tau h}^2$ is generally unidentified: observing only one potential outcome per unit does not reveal the within-block variance of individual treatment effects. A usable conservative estimator drops these nonnegative terms and substitutes the observed within-block sample variances s_{zh}^2 for S_{zh}^2:

$$\widehat{V}_{\text{Neyman}} = \sum_{h=1}^{H} \left(\frac{N_h}{N}\right)^2 \left(\frac{s_{1h}^2}{N_{1h}} + \frac{s_{0h}^2}{N_{0h}}\right).$$

This plug-in estimator requires enough observed units to estimate both within-block arm variances. In a matched-pair design there is only one observed outcome per arm in each pair, so uncertainty must instead be estimated from the collection of randomized pair contributions.

Blocking helps when the blocks capture pretreatment differences that predict outcomes. For example, if outcomes contain a large block-specific baseline component, that component is shared by the treated and control units compared within the block and cancels from their contrast. Between-block baseline differences therefore do not inflate the treatment comparison. Blocking also fixes arm counts within every block. With a common treatment fraction, it exactly balances the categorical block indicators; continuous covariates used to construct the blocks are balanced only approximately, to the extent that units placed together are similar.

Blocking is not free. Many fine blocks can be operationally difficult; attrition can break their intended structure; and a block must contain enough units to support the planned arm counts.

Blocking on weakly prognostic variables can provide little gain. The design should also be preserved in the analysis: estimate effects within blocks and aggregate using the target-population weights.

2.1 Matched pairs

Matched pairs are the limiting case $N_h = 2$ and $N_{1h} = 1$. Suppose there are $H = N/2$ pairs, and label the two units in pair h by a and b. Let $A_h = 1$ when unit a is treated and $A_h = 0$ when unit b is treated. The signed observed difference

$$D_h = (2A_h - 1)\,(Y_{ha}^{\text{obs}} - Y_{hb}^{\text{obs}})$$

always means "treated outcome minus control outcome": when $A_h = 1$, it is $Y_{ha}^{\text{obs}} - Y_{hb}^{\text{obs}}$; when $A_h = 0$, the sign reverses. Its design expectation is the average of the two unit-level effects in pair h. Consequently,

$$\hat{\tau}_{\text{pair}} = \frac{1}{H}\sum_{h=1}^{H} D_h$$

is design-unbiased for the finite-population ATE. Because each pair supplies only one treated and one control observation, there is no within-arm variance estimate inside a pair. Precision is instead learned from how the randomized contributions D_h vary across pairs.

Pairing can be powerful because assignment only decides which of two deliberately similar units is treated. If the matched units have similar potential outcomes under each arm, the two possible values of D_h are close, reducing its assignment variance. Pairing concentrates randomization locally: each pair has only two possible assignments, although assignments remain independent across pairs. Bai (2022) characterizes precision-optimal matched-pair designs, while Bai et al. (2022) develops inference that respects the pairing. Poor matches can lock in bad comparisons, and missing outcomes are especially damaging. Matching should therefore be based on pretreatment information chosen for prognostic value, not on an urge to balance every measured variable.

3 Design balance versus analysis adjustment

Return to the prior-achievement example. We can use the baseline score **before assignment**, by blocking students with similar scores, or **after assignment**, by statistically correcting whatever baseline-score difference happened to arise. These are different interventions on the experiment. The first changes the assignment distribution; the second leaves the design alone and changes the estimator. The broader design-first principle is articulated by Rubin (2008), while Bruhn and McKenzie (2009) compares common randomization procedures in field-experiment practice:

- **At design:** restrict or structure assignments so covariate imbalance is small by construction.
- **At analysis:** allow a simpler assignment design and subtract the part of the outcome difference predicted by realized covariate imbalance.

To make the analysis-stage option precise, suppose X_i is a column vector of pretreatment covariates, $\overline{X}_z^{\text{obs}}$ is its observed sample mean in arm z, and β is a coefficient vector of the same dimension. Under two-arm complete randomization, consider the regression-adjusted form

$$\hat{\tau}_{\text{adj}}(\beta) = \hat{\tau}_{\text{DM}} - \beta^\top(\overline{X}_1^{\text{obs}} - \overline{X}_0^{\text{obs}}).$$

The vector $\overline{X}_1^{\text{obs}} - \overline{X}_0^{\text{obs}}$ is the realized covariate imbalance. Its inner product with β is the part of the raw outcome difference attributed to that imbalance, so the estimator subtracts it. If β is fixed with respect to assignment, the correction has design expectation zero because complete randomization gives each arm the same expected covariate mean. A coefficient aligned with the outcome–covariate relationship can therefore reduce assignment variance without changing the design expectation; a poorly chosen coefficient can instead increase variance.

A coefficient fitted from the experiment's outcomes is assignment-dependent and needs the finite-sample or asymptotic justification discussed in L02 (Freedman 2008; Lin 2013). Design balance is visible before outcomes are observed and protects a range of later analyses. Analysis adjustment can use continuous, high-dimensional, or nonlinear predictions more flexibly, but it requires appropriate estimation and uncertainty calculations. The two are complements: one can block on a few decisive variables and adjust for remaining prognostic variation.

This comparison also reveals a general principle. Design validity does not require a correct outcome model relating Y to X; its precision gains do. Balancing weak or wrongly represented features may provide little benefit, while concentrating the design too aggressively around one criterion can reduce robustness to outcome structure that criterion missed.

4 AIPW: prediction plus randomized correction

Covariate adjustment presents a useful tension. An outcome model can predict a missing potential outcome for every unit, but a misspecified model need not average to the correct finite-population mean. An HT estimator has the opposite strength: known assignment probabilities make it design-unbiased, but it may be noisy because it inverse-weights the full observed outcomes. **Augmented inverse probability weighting** (AIPW) combines these two starting points. It uses predictions to remove explainable variation and uses randomization to correct the predictions rather than trusting them outright.

The target for arm z is the finite-population mean $\overline{Y}(z)$. Let $m_z(x)$ predict the potential outcome under arm z from pretreatment covariates. For the moment, treat the N numbers $m_z(X_i)$ as fixed before assignment; they need not equal the actual potential outcomes. Also assume the relevant positivity condition

$$\pi_i(z) = \mathbb{P}_Z(Z_i = z) > 0 \qquad \text{for every unit and arm under study.}$$

Here $\pi_i(z)$ is the known probability that the design assigns unit i to arm z. Positivity ensures that every unit whose potential outcome contributes to $\overline{Y}(z)$ has some chance of revealing that outcome.

The population average of the predictions, $N^{-1}\sum_i m_z(X_i)$, is available because the covariates are observed for all experimental units. Correct it using the prediction errors observed in arm z:

$$\hat{\mu}_{z,\text{AIPW}} = \frac{1}{N}\sum_{i=1}^{N}\left[m_z(X_i) + \frac{\mathbf{1}\{Z_i = z\}}{\pi_i(z)}\{Y_i^{\text{obs}} - m_z(X_i)\}\right], \tag{5}$$

The indicator $\mathbf{1}\{Z_i = z\}$ is one exactly when unit i is assigned to arm z. Thus the second term is zero for units outside that arm and equals the observed residual divided by its assignment probability for units inside it. Subtracting the two arm-specific mean estimators gives

$$\hat{\tau}_{\text{AIPW}} = \hat{\mu}_{1,\text{AIPW}} - \hat{\mu}_{0,\text{AIPW}}$$

as an estimator of $\tau = \overline{Y}(1) - \overline{Y}(0)$ (Robins et al. 1994; Tsiatis et al. 2008). The GREG construction in L02 has the same prediction-plus-weighted-residual architecture; AIPW writes it arm by arm and allows general prediction functions.

The construction can be read in three steps:

1. predict every unit's potential outcome under arm z;
2. among units actually observed in arm z, inverse-weight the prediction error;
3. add that correction to the population average prediction.

The correction works because, for each unit separately, inverse-probability weighting recovers the full prediction error. Since $Y_i^{\text{obs}} = Y_i(z)$ on the event $Z_i = z$,

$$\mathbb{E}_Z\left[\frac{\mathbf{1}\{Z_i = z\}}{\pi_i(z)}\{Y_i^{\text{obs}} - m_z(X_i)\}\right] = Y_i(z) - m_z(X_i).$$

Adding the fixed prediction back therefore gives design unbiasedness without assuming that the prediction is correct:

$$\begin{aligned}\mathbb{E}_Z(\hat{\mu}_{z,\text{AIPW}}) &= \frac{1}{N}\sum_i \left[m_z(X_i) + Y_i(z) - m_z(X_i)\right] \\ &= \overline{Y}(z).\end{aligned}$$

This is the central division of labor. The model supplies a precision device; the randomization correction supplies validity for the finite-population target. When $m_z(X_i)$ is close to $Y_i(z)$, the residuals passed through the random inverse-probability weights are smaller and the estimator can be more precise than HT. A poor fixed predictor does not destroy design unbiasedness, though it can increase variance.

> ⚠ Fitted predictions are part of the estimator
>
> The exact argument above treats m_z as fixed with respect to assignment. An externally trained or otherwise preassignment predictor preserves that exact conditional design-unbiasedness. If the same randomized outcomes are used to fit m_z, exact finite-sample unbiasedness does not follow automatically. Under independent assignment, carefully constructed leave-one-out or

cross-fitted predictions can separate a unit's assignment from its prediction. Under complete, blocked, or rerandomized designs, assignments are dependent, so cross-fitting alone is not a universal finite-sample fix; one needs a design-specific argument or suitable asymptotics. Randomization remains the foundation, but "the model may be wrong" does not mean "the fitting procedure is irrelevant."

In observational causal inference, AIPW is often introduced through double robustness. That is not the focus here: the assignment probabilities in a randomized experiment are known. Our emphasis is the control-variate interpretation—prediction for precision, inverse-probability correction for the design-based target.

5 Pair matching in cluster-randomized experiments

Sometimes treatment is assigned to schools, villages, or clinics while outcomes are measured on individuals. With few clusters, ordinary cluster randomization can leave large baseline differences. Imai et al. (2009) argue for pairing similar clusters and randomizing one cluster within each pair; Middleton and Aronow (2015) develops the design-based estimand and weighting issues for cluster-randomized experiments.

The distinction between the unit of assignment and the unit of measurement is essential. Suppose there are K pairs indexed by $k = 1, \ldots, K$, and pair k contains clusters $j = 1, 2$. Cluster (k, j) contains n_{kj} individuals, and $N = \sum_{k=1}^{K} \sum_{j=1}^{2} n_{kj}$ is the total number of individuals. Under cluster-level treatment and no interference between clusters, define the cluster potential-outcome total

$$T_{kj}(z) = \sum_{\ell=1}^{n_{kj}} Y_{kj\ell}(z),$$

where ℓ indexes individuals within a cluster. The individual-weighted finite-population ATE is then

$$\tau = \frac{1}{N} \sum_{k=1}^{K} \sum_{j=1}^{2} \{T_{kj}(1) - T_{kj}(0)\}.$$

Let $A_k = 1$ mean that cluster 1 in pair k is treated; when $A_k = 0$, cluster 2 is treated. The two possibilities occur with probability $1/2$, independently across pairs. Write T_{kj}^{obs} for the total under the assignment cluster (k, j) actually receives. The HT estimator of the individual-weighted ATE is

$$\hat{\tau}_{\text{HT}} = \frac{2}{N} \sum_{k=1}^{K} (2A_k - 1)\{T_{k1}^{\text{obs}} - T_{k2}^{\text{obs}}\}.$$

The sign $2A_k - 1$ always orients the expression as treated-cluster total minus control-cluster total. The leading factor 2 is the inverse of each cluster's treatment or control probability. Averaging over the two assignments in pair k recovers the sum of the treatment effects for all individuals in both clusters; dividing by N therefore targets the individual-weighted ATE above.

Each pair contributes one independent randomized contrast, so design-based uncertainty is built from variation across those pair contributions. Two details are easy to miss:

- The effective randomization units are clusters, not the individuals nested within them.
- Weighting determines the estimand. The total-based HT expression targets an average over individuals. Averaging cluster means equally instead targets a cluster-weighted or pair-weighted effect; unequal cluster sizes make the distinction consequential.

Pair matching can substantially improve precision when matches are prognostic, but the analysis must honor both pairing and unequal cluster sizes. Treating all individual observations as independently randomized produces fictitious precision.

6 Rerandomization

Blocking is attractive when units fall naturally into small groups within which fixed treatment counts can be enforced. With several continuous covariates, however, useful blocks may be difficult to define. **Rerandomization** takes a different route: repeatedly draw an assignment from a familiar proposal design, measure its global covariate imbalance, and retain the draw only if that imbalance meets a prespecified standard (Morgan and Rubin 2012). It changes the design before outcomes are seen; it is not post hoc deletion of an inconvenient randomization.

Write $\mathbb{P}_{\mathrm{CR}}$ for the proposal complete-randomization design and assume equal-sized treatment arms. For the pretreatment covariate vector X_i, let

$$\hat{\tau}_X = \overline{X}_1^{\mathrm{obs}} - \overline{X}_0^{\mathrm{obs}}$$

be the vector of treated-minus-control covariate mean differences. Zero is perfect mean balance. Different coordinates may have different units and may be strongly correlated, so an ordinary Euclidean norm is not a meaningful common scale. Assuming the covariance matrix is nonsingular, the Mahalanobis balance score (Mahalanobis 1936) standardizes the imbalance by its covariance under the proposal design:

$$M(\mathbf{Z}) = \hat{\tau}_X^{\top} \left\{ \mathrm{Cov}_{\mathrm{CR}}(\hat{\tau}_X) \right\}^{-1} \hat{\tau}_X. \tag{6}$$

The covariance subscript "CR" matters: $\mathrm{Cov}_{\mathrm{CR}}(\hat{\tau}_X)$ describes how covariate imbalance would vary over repeated proposal assignments, with the covariates held fixed. Consequently, $M(\mathbf{Z})$ is a scalar standardized squared distance from exact balance. With a single covariate, it is simply the squared mean difference divided by its complete-randomization variance.

Choose a threshold a before assignment so that the acceptance event has positive proposal probability. Repeatedly draw a complete randomization and accept the first draw satisfying $M(\mathbf{Z}) \leq a$. The actual rerandomization design, denoted $\mathbb{P}_{\mathrm{ReM}}$, is the proposal distribution conditional on acceptance:

$$\begin{aligned} \mathbb{P}_{\mathrm{ReM}}(\mathbf{Z} = \mathbf{z}) &= \mathbb{P}_{\mathrm{CR}}\{\mathbf{Z} = \mathbf{z} \mid M(\mathbf{Z}) \leq a\} \\ &= \frac{\mathbb{P}_{\mathrm{CR}}(\mathbf{Z} = \mathbf{z}) \mathbf{1}\{M(\mathbf{z}) \leq a\}}{\mathbb{P}_{\mathrm{CR}}\{M(\mathbf{Z}) \leq a\}}. \end{aligned} \tag{7}$$

If complete randomization is uniform over assignments with the prescribed arm sizes, rerandomization is uniform over the subset that passes the balance rule. This is still a randomized design, provided the covariates, score, and threshold are fixed in advance and the acceptance procedure is recorded. Ad hoc rejection until the groups merely "look nice" does not define a reproducible assignment law.

6.1 What rerandomization buys

Let $\mathbf{1}_N$ denote the length-N vector of ones, so the treatment-control complement of $\mathbf{z}$ is $\mathbf{1}_N - \mathbf{z}$. If the acceptance rule is symmetric—$\mathbf{z}$ is accepted exactly when its complement is—then every accepted assignment has an equally likely accepted partner that reverses all arm labels. With equal arm sizes, each unit therefore retains marginal treatment probability 1/2, and difference in means remains design-unbiased. The joint assignment distribution has nevertheless changed: assignments with large $M(\mathbf{Z})$ now have probability zero. When the balanced covariates predict outcomes, removing those assignments can substantially reduce variance.

But the analysis must use the design that was actually run. An exact Fisher randomization test of a sharp null should sample from accepted assignments, not from all complete randomizations. Under the regularity conditions studied by Li et al. (2018), the usual Neyman normal interval calculated as if complete randomization had been used is asymptotically conservative; this is not a universal finite-sample statement.

The large-sample result of Li et al. (2018) explains the mechanism. Decompose the outcome variation into a component linearly associated with the balanced covariates and a residual component orthogonal to them. Under their regularity conditions, the difference-in-means fluctuation likewise separates into two independent pieces. The residual piece remains asymptotically Gaussian because the acceptance rule does not directly constrain it. The covariate-associated piece has a Gaussian limit truncated to the accepted balance region. Rerandomization therefore shrinks precisely the part of the estimator's variation aligned with X. The precision gain is larger when that part explains more of the outcome and when the threshold is more stringent. Li and Ding (2020) then studies how regression adjustment and rerandomization interact rather than treating design- and analysis-stage balance as substitutes.

6.2 The robustness tradeoff

Rerandomization has an appealing operational story and preserves multiple acceptable assignments. Its weaknesses are equally instructive:

- a stringent threshold lowers the acceptance probability and concentrates the assignment distribution;
- Mahalanobis balance is only as relevant as the chosen covariates and feature representation; in high dimensions, covariance regularization also changes the metric;
- for a fixed acceptance probability, increasing dimension weakens the attainable improvement in each direction and makes covariance estimation or regularization more consequential;
- unrecorded changes to the acceptance rule invalidate the stated design;
- analysis that pretends complete randomization was used mischaracterizes uncertainty.

The threshold therefore mediates balance against robustness. Making a smaller improves balance along the chosen covariate directions, but it leaves fewer possible assignments and makes the design more dependent on whether those directions capture the outcome structure that matters. Here robustness includes both protection against a misspecified balance criterion and enough randomness to support credible, reproducible inference. Perfect observed balance is not the only goal of a randomized experiment.

7 What to carry forward

Experimental design controls the dependence among assignments. Blocking uses scientifically meaningful local comparisons; matching makes those blocks as small as possible; rerandomization conditions on acceptable global balance. Analysis adjustment and AIPW can recover additional precision after a simpler design. In every case, the estimator and its uncertainty calculation must correspond to the assignment distribution that was actually used.

8 Advanced material

8.1 Gram–Schmidt walk designs

The word **discrepancy** describes a signed imbalance. To see the connection to experiments, suppose the treatment and control arms have equal size and define the assignment sign $s_i = 2Z_i - 1$, which is $+1$ for treatment and -1 for control. Then

$$\hat{\tau}_X = \overline{X}_1^{\text{obs}} - \overline{X}_0^{\text{obs}} = \frac{2}{N} \sum_{i=1}^{N} s_i X_i.$$

Thus choosing signs that make the vector sum $\sum_i s_i X_i$ small is exactly choosing an assignment with small covariate mean imbalance. Discrepancy theory studies how signs can be chosen so that many coordinates or other linear summaries remain small simultaneously. The norm used to judge the signed sum specifies which kinds of imbalance matter.

A purely discrepancy-minimizing algorithm might return one deterministic assignment. The Gram–Schmidt walk (GSW) instead turns a discrepancy-control procedure into a randomized assignment design (Harshaw et al. 2024). It generates the signs through a random walk while steering their covariate sum toward balance.

The design has a tuning parameter that governs how aggressively the walk prioritizes covariate balance relative to unconstrained randomization. At one extreme it behaves more like conventional random assignment; at the other it pushes harder to keep the linear discrepancy small. Crucially, it retains prescribed unit-level marginals, such as $\mathbb{P}_Z(Z_i = 1) = 1/2$, while inducing dependence among assignments. The resulting guarantees concern design-based mean squared error over specified classes of outcome structure. This separates GSW from selecting a single best-looking assignment: its balance comes from a probability distribution over assignments, not from abandoning randomization.

The guarantee also states its own limitation: linear balance protects outcomes whose conditional mean is well represented by the chosen linear features. Cytrynbaum (2026) shows that uniformly approaching classical efficiency over broad smooth outcome classes becomes impossible once dimension grows too quickly. The paper develops a nonparametric, kernelized GSW in a reproducing-kernel Hilbert space and identifies more favorable rates under additive structure. Kernelization expands the function class being balanced; it does not eliminate the need to say which structure is plausible.

8.2 Correlated cluster designs without rich covariates

The paired-cluster discussion used covariates to decide which clusters should be compared. Candogan et al. (2024) ask a different question: can dependence be designed robustly when cluster size is the principal available structure? They study correlated cluster assignments with equal marginal treatment probabilities and a cluster-aware HT estimator, choosing dependence to minimize a worst-case variance criterion.

Their imbalance-reducing construction uses cluster sizes rather than a rich outcome-prediction model. The asymptotic guarantees require regularity, including that no single cluster dominate the population. The result is therefore neither covariate matching nor a claim that size is always the most prognostic variable. It shows that assignment correlation can be optimized against a stated worst-case class even when conventional matching information is sparse.

8.3 Optimization-based designs and the value of randomness

Once discrepancy has been written as a function of the assignment signs, one can search directly for the assignment minimizing it—for example, minimizing a norm of $\sum_i s_i X_i$ subject to fixed arm sizes. Mixed-integer optimization can produce striking balance in small samples (Bertsimas et al. 2015), and minimax formulations connect classes of plausible outcome functions to the imbalance metric that should be optimized (Kallus 2018). The objective is never neutral: choosing a norm, feature map, or function class says which outcome patterns the design is intended to protect.

Selecting one minimizing assignment eliminates design-based randomization. Randomizing between that assignment and its treatment-control complement restores unit-level marginals of $1/2$, but leaves only two assignments in the support. Either construction can be brittle to aspects of the outcome surface omitted from the objective. More diffuse randomization distributes that risk. Kallus (2021) formalizes this through mixed-strategy optimal designs: with no informative structure, complete randomization is minimax; gains require restrictions on plausible outcome functions, and inference supplies a reason to retain sufficiently diffuse randomization.

GSW, rerandomization, and mixed-strategy optimization occupy different points on the same frontier. Each asks how much imbalance to remove while preserving enough randomness for robustness and inference.

8.4 Efron's biased coin

Efron's biased-coin design addresses a sequential problem: how can an experiment resist persistent arm-size imbalance without making its next assignment completely predictable? (Efron 1971) After n assignments, let $D_n = N_{1n} - N_{0n}$ be the treated count minus the control count. If $D_n = 0$, assign

the next unit by a fair coin. If $D_n > 0$, treatment is overrepresented, so assign the next unit to control with a chosen probability $q > 1/2$; if $D_n < 0$, assign it to treatment with that probability. The parameter q controls the pull back toward equal counts.

At $q = 1/2$, the feedback disappears and the rule is ordinary i.i.d. Bernoulli assignment. As q rises toward one, balance is restored more forcefully but future assignments become easier to predict. The design therefore makes the balance–randomness tradeoff explicit. Covariate-adaptive procedures such as Pocock and Simon (1975) generalize the same feedback idea to prognostic-factor imbalance, but the dependence created by that feedback must be reflected in the analysis.

8.5 D-, A-, and E-optimal design

Section 7.5 of Boyd and Vandenberghe (2004) gives a modern convex-optimization treatment of a classical model-based design problem whose general theory traces to Kiefer (1959). Suppose measurement option r has a known regressor vector v_r, and let $\lambda_r \geq 0$ be the fraction of measurement effort allocated to that option, with $\sum_r \lambda_r = 1$. In a homoskedastic linear model, the normalized information matrix is

$$\mathcal{I}(\lambda) = \sum_r \lambda_r v_r v_r^\top.$$

When this matrix is nonsingular, the coefficient covariance is proportional to $\mathcal{I}(\lambda)^{-1}$. The design problem chooses the allocation λ to make that covariance small, but a matrix is not ordered by a single universally best scalar. Common criteria emphasize different geometries:

- **D-optimality** maximizes $\det\{\mathcal{I}(\lambda)\}$, equivalently shrinking the volume of the coefficient confidence ellipsoid.
- **A-optimality** minimizes $\mathrm{tr}\{\mathcal{I}(\lambda)^{-1}\}$, the sum of the marginal coefficient-variance proxies.
- **E-optimality** maximizes the smallest eigenvalue of $\mathcal{I}(\lambda)$, protecting the least-informed coefficient direction.

These criteria are connected to covariate-balanced treatment design through convex optimization and worst-direction reasoning. They are not the same problem. Classical optimal design begins from a response model and chooses where to measure parameters; GSW chooses a randomized treatment allocation and seeks design-based protection over outcome classes. The analogy is most useful when it makes the assumed geometry explicit.

8.6 Staggered rollouts beyond SUTVA

In a staggered rollout, clusters or units adopt treatment at different assigned times. The stepped-wedge formulation of Hussey and Hughes (2007) is a canonical example. The design creates comparisons across adoption cohorts and calendar periods. Operational constraints often motivate it, but the schedule can also generate deliberate variation in treatment saturation and exposure duration.

That variation is valuable when no-interference or no-carryover assumptions are implausible: outcomes can be indexed by neighbors' adoption, local saturation, or time since treatment rather than by a single binary label. The cost is that calendar-time trends, anticipation, and dynamic effects

become part of the estimand and design. A rollout is therefore not merely a delayed two-arm experiment; it requires a potential-outcome table indexed by assignment histories.

9 Further reading

For historical orientation, Fisher (1935) develops randomization and blocking, while Rubin (2008) states the modern design-before-analysis principle. Ding (2023) provides a concise contemporary treatment of finite-population experiments and rerandomization. For matched designs, read Bai (2022) and Bai et al. (2022) alongside the cluster-randomized perspective of Imai et al. (2009) and Middleton and Aronow (2015).

For rerandomization, begin with Morgan and Rubin (2012) for the design principle, then read Li et al. (2018) for the non-Gaussian asymptotic distribution induced by acceptance and Li and Ding (2020) for its interaction with regression adjustment. Candogan et al. (2024) shows how cluster-assignment dependence can instead be optimized from cluster-size information and a worst-case criterion.

The modern balance–robustness sequence runs from optimization-based designs (Bertsimas et al. 2015; Kallus 2018, 2021) to randomized discrepancy control (Harshaw et al. 2024) and its recent nonlinear extension (Cytrynbaum 2026). Efron (1971) and Pocock and Simon (1975) provide sequential counterparts: use randomization feedback to resist arm-size or covariate imbalance without becoming deterministic. For classical information-matrix optimality, pair Kiefer (1959) with the computational treatment in Boyd and Vandenberghe (2004); for staggered cluster rollouts, see Hussey and Hughes (2007).

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References

Bai, Yuehao. 2022. "Optimality of Matched-Pair Designs in Randomized Controlled Trials." *American Economic Review* 112 (12): 3911–40. https://doi.org/10.1257/aer.20201856.

Bai, Yuehao, Joseph P. Romano, and Azeem M. Shaikh. 2022. "Inference in Experiments with Matched Pairs." *Journal of the American Statistical Association* 117 (540): 1726–37. https://doi.org/10.1080/01621459.2021.1883437.

Bertsimas, Dimitris, Mac Johnson, and Nathan Kallus. 2015. "The Power of Optimization over Randomization in Designing Experiments Involving Small Samples." *Operations Research* 63 (4): 868–76. https://doi.org/10.1287/opre.2015.1361.

Boyd, Stephen, and Lieven Vandenberghe. 2004. *Convex Optimization*. Cambridge University Press. https://doi.org/10.1017/CBO9780511804441.

Bruhn, Miriam, and David McKenzie. 2009. "In Pursuit of Balance: Randomization in Practice in Development Field Experiments." *American Economic Journal: Applied Economics* 1 (4): 200–232. https://doi.org/10.1257/app.1.4.200.

Candogan, Ozan, Chen Chen, and Rad Niazadeh. 2024. "Correlated Cluster-Based Randomized Experiments: Robust Variance Minimization." *Management Science* 70 (6): 4069–86. https://doi.org/10.1287/mnsc.2021.02741.

Cytrynbaum, Max. 2026. *The Limits of Experimental Design: Covariate Balance Beyond Low Dimension*. https://doi.org/10.48550/arXiv.2608.18057.

Ding, Peng. 2023. *A First Course in Causal Inference*. https://doi.org/10.48550/arXiv.2305.18793.

Efron, Bradley. 1971. "Forcing a Sequential Experiment to Be Balanced." *Biometrika* 58 (3): 403–17. https://doi.org/10.1093/biomet/58.3.403.

Fisher, Ronald A. 1935. *The Design of Experiments*. Oliver & Boyd.

Freedman, David A. 2008. "On Regression Adjustments to Experimental Data." *Advances in Applied Mathematics* 40 (2): 180–93. https://doi.org/10.1016/j.aam.2006.12.003.

Harshaw, Christopher, Fredrik Sävje, Daniel A. Spielman, and Peng Zhang. 2024. "Balancing Covariates in Randomized Experiments with the Gram–Schmidt Walk Design." *Journal of the American Statistical Association* 119 (548): 2934–46. https://doi.org/10.1080/01621459.2023.2285474.

Horvitz, D. G., and D. J. Thompson. 1952. "A Generalization of Sampling Without Replacement from a Finite Universe." *Journal of the American Statistical Association* 47 (260): 663–85. https://doi.org/10.1080/01621459.1952.10483446.

Hussey, Michael A., and James P. Hughes. 2007. "Design and Analysis of Stepped Wedge Cluster Randomized Trials." *Contemporary Clinical Trials* 28 (2): 182–91. https://doi.org/10.1016/j.cct.2006.05.007.

Imai, Kosuke, Gary King, and Clayton Nall. 2009. "The Essential Role of Pair Matching in Cluster-Randomized Experiments, with Application to the Mexican Universal Health Insurance Evaluation." *Statistical Science* 24 (1): 29–53. https://doi.org/10.1214/08-STS274.

Kallus, Nathan. 2018. "Optimal a Priori Balance in the Design of Controlled Experiments." *Journal of the Royal Statistical Society: Series B* 80 (1): 85–112. https://doi.org/10.1111/rssb.12240.

Kallus, Nathan. 2021. "On the Optimality of Randomization in Experimental Design: How to Randomize for Minimax Variance and Design-Based Inference." *Journal of the Royal Statistical Society: Series B* 83 (2): 404–9. https://doi.org/10.1111/rssb.12412.

Kiefer, Jack. 1959. "Optimum Experimental Designs." *Journal of the Royal Statistical Society: Series B* 21 (2): 272–304. https://doi.org/10.1111/j.2517-6161.1959.tb00338.x.

Li, Xinran, and Peng Ding. 2020. "Rerandomization and Regression Adjustment." *Journal of the Royal Statistical Society: Series B* 82 (1): 241–68. https://doi.org/10.1111/rssb.12353.

Li, Xinran, Peng Ding, and Donald B. Rubin. 2018. "Asymptotic Theory of Rerandomization in

Treatment–Control Experiments." *Proceedings of the National Academy of Sciences* 115 (37): 9157–62. https://doi.org/10.1073/pnas.1808191115.

Lin, Winston. 2013. "Agnostic Notes on Regression Adjustments to Experimental Data: Reexamining Freedman's Critique." *The Annals of Applied Statistics* 7 (1): 295–318. https://doi.org/10.1214/12-AOAS583.

Mahalanobis, Prasanta Chandra. 1936. "On the Generalised Distance in Statistics." *Proceedings of the National Institute of Sciences of India* 2 (1): 49–55.

Middleton, Joel A., and Peter M. Aronow. 2015. "Unbiased Estimation of the Average Treatment Effect in Cluster-Randomized Experiments." *Statistics, Politics and Policy* 6 (1-2): 39–75. https://doi.org/10.1515/spp-2013-0002.

Morgan, Kari Lock, and Donald B. Rubin. 2012. "Rerandomization to Improve Covariate Balance in Experiments." *The Annals of Statistics* 40 (2): 1263–82. https://doi.org/10.1214/12-AOS1008.

Pocock, Stuart J., and Richard Simon. 1975. "Sequential Treatment Assignment with Balancing for Prognostic Factors in the Controlled Clinical Trial." *Biometrics* 31 (1): 103–15. https://doi.org/10.2307/2529712.

Robins, James M., Andrea Rotnitzky, and Lue Ping Zhao. 1994. "Estimation of Regression Coefficients When Some Regressors Are Not Always Observed." *Journal of the American Statistical Association* 89 (427): 846–66. https://doi.org/10.1080/01621459.1994.10476818.

Rubin, Donald B. 2008. "For Objective Causal Inference, Design Trumps Analysis." *The Annals of Applied Statistics* 2 (3): 808–40. https://doi.org/10.1214/08-AOAS187.

Splawa-Neyman, Jerzy, Dorota M. Dabrowska, and Terence P. Speed. 1990. "On the Application of Probability Theory to Agricultural Experiments: Essay on Principles, Section 9." *Statistical Science* 5 (4): 465–72. https://doi.org/10.1214/ss/1177012031.

Tsiatis, Anastasios A., Marie Davidian, Min Zhang, and Xiaomin Lu. 2008. "Covariate Adjustment for Two-Sample Treatment Comparisons in Randomized Clinical Trials: A Principled yet Flexible Approach." *Statistics in Medicine* 27 (23): 4658–77. https://doi.org/10.1002/sim.3113.